

Monocular Bootstrapped Panoptic Segmentation: Unsupervised Partitioning of Images into Semantic Regions and Object Instances without Manual Annotation

Abstract

Unsupervised panoptic segmentation seeks to partition images into semantic regions and object instances without any manual annotation. Existing approaches rely on stereo depth cues or optical flow for motion segmentation, limiting their applicability to monocular settings. We present a fully monocular pipeline for unsupervised panoptic segmentation that achieves state-of-the-art performance on Cityscapes without requiring stereo cameras, LiDAR, or any form of human supervision.

Our method proceeds in three stages. In the first stage, a Depth-Conditioned Feature Adapter fuses semantic representation vectors with monocular depth encodings through a lightweight multi-layer perceptron. The adapter employs zero-initialized output weights and a feature preservation loss to ensure stable training. In the second stage, depth-conditioned representations are clustered into semantic categories and instance masks are extracted through contour-aware instance splitting. In the third stage, the resulting pseudo-labels supervise a panoptic segmentation model through exponential moving average self-training.

On Cityscapes validation, our pipeline achieves a panoptic quality of 35.83%, establishing a new state-of-the-art for fully monocular unsupervised panoptic segmentation and exceeding the stereo-dependent prior state-of-the-art by 8.0 absolute points. On cross-dataset transfer to Mapillary Vistas and KITTI, the method achieves 39.19% and 34.85% panoptic quality respectively, demonstrating strong generalization across diverse driving scenarios.

1 Introduction

1.1 Motivation and Problem Statement

Panoptic segmentation unifies semantic segmentation and instance segmentation into a single coherent task, assigning every pixel in an image both a semantic label and, for thing classes, an instance identifier. The dominant paradigm for achieving this capability has been fully supervised learning on large datasets annotated by human annotators. However, the annotation cost for panoptic labels is prohibitive: a single Cityscapes image requires approximately 1.5 hours of expert annotator time for panoptic annotation. This cost scales linearly with dataset size, creating a fundamental bottleneck for scaling panoptic segmentation to new domains.

Unsupervised panoptic segmentation offers an alternative path: partition images into semantic regions and object instances without any manual annotation. The challenge is formidable because the task requires simultaneously solving two ill-posed problems, discovering coherent semantic categories and distinguishing individual object instances, from unlabeled data alone.

1.2 Limitations of Prior Approaches

Prior work on unsupervised panoptic segmentation has relied heavily on geometric cues from stereo cameras. CUPS, the current state-of-the-art, requires calibrated stereo pairs to compute disparity maps, which are then used to guide semantic clustering and instance separation. This stereo dependency fundamentally limits applicability: stereo cameras are expensive, require careful calibration, and fail in textureless regions where disparity estimation becomes unreliable.

Monocular depth estimation offers a pathway to remove the stereo dependency, but introduces its own challenges. Monocular depth predictors are inherently scale-ambiguous and produce boundary-smoothed outputs that corrupt instance separation. The binding constraint in monocular unsupervised panoptic segmentation is not the quality of semantic clustering, but the precision of depth boundaries for instance splitting.

1.3 Contributions

This paper makes the following contributions. First, we present a fully monocular pipeline for unsupervised panoptic segmentation that achieves 35.83% panoptic quality on Cityscapes, exceeding the stereo-dependent state-of-the-art by 8.0 absolute points. Second, we introduce a Depth-Conditioned Feature Adapter that fuses semantic codes with monocular depth encodings through a lightweight architecture with 225,114 parameters, trained end-to-end with a feature preservation loss. Third, we demonstrate that the binding constraint in monocular unsupervised panoptic segmentation is depth boundary quality rather than semantic clustering accuracy, and we propose a contour-aware instance splitting module that addresses this limitation. Fourth, we show strong cross-dataset generalization to Mapillary Vistas and KITTI without any domain adaptation.

2 Related Work

2.1 Unsupervised Semantic Segmentation

Early approaches to unsupervised semantic segmentation relied on handcrafted features and clustering algorithms. More recent methods leverage self-supervised visual representations from foundation models such as DINO and DINOv2, which produce semantically meaningful feature maps without any manual annotation. CAUSE-TR demonstrated that semantic textural representations extracted from DINOv2 ViT features can be clustered into coherent semantic categories through k-means overclustering. However, these approaches stop at semantic segmentation and do not address instance separation.

2.2 Unsupervised Instance Segmentation

Unsupervised instance segmentation has progressed through two main paradigms: motion-based and depth-based approaches. Motion-based methods leverage optical flow or video sequences to segment moving objects, but fail on static scenes. Depth-based methods use stereo disparity or monocular depth to separate objects at different depth planes. The key challenge in depth-based instance segmentation is handling objects at similar depths, which requires additional cues such as contour detection or semantic boundaries.

2.3 Unsupervised Panoptic Segmentation

CUPS established the first complete pipeline for unsupervised panoptic segmentation, combining stereo depth with semantic clustering and motion segmentation. Their approach achieved 27.8% panoptic quality on Cityscapes, but required calibrated stereo pairs. Our work removes the stereo dependency entirely, replacing stereo disparity with monocular depth from a single camera.

2.4 Parameter-Efficient Adaptation

The literature on parameter-efficient fine-tuning has explored numerous adapter architectures including bottleneck adapters, prefix tuning, and prompt tuning. For our depth-semantic fusion module, we employ a custom multi-layer perceptron with zero-initialized output weights, drawing inspiration from the general principle that adapters should start from identity and learn deviations. This design choice is distinct from low-rank adaptation methods, as our fusion module operates on cross-modal concatenated features rather than reparameterizing pre-trained weight matrices.

3 Methodology

Our pipeline proceeds through three stages. Stage 1 produces depth-conditioned semantic representations. Stage 2 generates pseudo-labels through semantic clustering and instance splitting. Stage 3 trains a panoptic segmentation model via self-training. We describe each stage in detail, with exact mathematical formulations derived from our implementation.

3.1 Stage 1: Depth-Conditioned Feature Adaptation

3.1.1 Input Representations

Stage 1 accepts two inputs per image patch. The first input is a semantic code vector extracted from the CAUSE-TR clustering head. For each patch i , the semantic code is denoted $f_i \in \mathbb{R}^{90}$. These codes are produced by a projection head applied to DINOv2 ViT-B/14 patch tokens, followed by k-means clustering with $k = 80$ centroids. The 80 clusters are then mapped to 19 Cityscapes thing classes and 8 stuff classes through majority voting, yielding 90-dimensional representation vectors.

The second input is a depth encoding vector. For each patch i , we compute a monocular depth estimate using DepthPro, producing a single scalar depth value. This scalar is encoded into a 16-dimensional vector $d_i \in \mathbb{R}^{16}$ through sinusoidal positional encoding with logarithmically spaced frequency bands:

$$d_i^{(j)} = \sin\left(\frac{z_i}{\tau_j}\right) \quad \text{for } j = 1, \dots, 8, \quad d_i^{(j+8)} = \cos\left(\frac{z_i}{\tau_j}\right) \quad \text{for } j = 1, \dots, 8, \quad (1)$$

where z_i is the monocular depth estimate for patch i and τ_j are exponentially spaced temperature parameters.

3.1.2 Adapter Architecture

The two input vectors are concatenated to form a 106-dimensional vector $x_i = [f_i; d_i] \in \mathbb{R}^{106}$. This vector is processed by a two-layer multi-layer perceptron with hidden dimension 384. The first linear transformation maps from 106 to 384 dimensions:

$$h_i^{(1)} = W_1 x_i + b_1, \quad W_1 \in \mathbb{R}^{384 \times 106}, \quad b_1 \in \mathbb{R}^{384}. \quad (2)$$

The hidden representation undergoes layer normalization and a rectified linear unit activation:

$$a_i^{(1)} = \text{ReLU}\left(\text{LayerNorm}(h_i^{(1)})\right). \quad (3)$$

The second hidden layer maps from 384 to 384 dimensions:

$$h_i^{(2)} = W_2 a_i^{(1)} + b_2, \quad W_2 \in \mathbb{R}^{384 \times 384}, \quad b_2 \in \mathbb{R}^{384}, \quad (4)$$

followed by another layer normalization and ReLU activation:

$$a_i^{(2)} = \text{ReLU}\left(\text{LayerNorm}(h_i^{(2)})\right). \quad (5)$$

The output layer maps from 384 to 90 dimensions:

$$\Delta f_i = W_3 a_i^{(2)} + b_3, \quad W_3 \in \mathbb{R}^{90 \times 384}, \quad b_3 \in \mathbb{R}^{90}. \quad (6)$$

The output weights W_3 and biases b_3 are initialized to zero, ensuring that at the start of training the adapter acts as an identity mapping: $\Delta f_i = 0$ and the depth-conditioned feature equals the original semantic code. The adapted feature is computed through a residual addition:

$$\tilde{f}_i = f_i + \Delta f_i. \quad (7)$$

The total parameter count of the adapter is 225,114 parameters, computed as $(106 \times 384 + 384) + (384 \times 384 + 384) + (384 \times 90 + 90) = 41,088 + 147,840 + 34,650 = 223,578$ for the linear layers plus $2 \times 384 \times 2 = 1,536$ for the layer normalization parameters.

3.1.3 Training Objective

The adapter is trained on the Cityscapes training set containing 2,975 images. The training objective consists of two terms. The first term is a depth correlation loss that encourages the adapted features to respect depth boundaries:

$$\mathcal{L}_{\text{depth-corr}} = \sum_{i,j} \left| \langle \tilde{f}_i - \tilde{f}_j, \tilde{f}_i - \tilde{f}_j \rangle - |z_i - z_j|^2 \right|, \quad (8)$$

where the sum is over pairs of patches within a local spatial neighborhood. This loss encourages the feature distance between patches to approximate their depth difference.

The second term is a feature preservation loss that prevents the adapter from deviating too far from the original semantic codes:

$$\mathcal{L}_{\text{preserve}} = \sum_i \|\Delta f_i\|_2^2 = \sum_i \|\tilde{f}_i - f_i\|_2^2. \quad (9)$$

The total training loss is a weighted combination:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{depth-corr}} + \lambda_{\text{preserve}} \cdot \mathcal{L}_{\text{preserve}}, \quad (10)$$

where the preservation weight is set to $\lambda_{\text{preserve}} = 20.0$ in all experiments. This high weight ensures that the adapter primarily refines rather than replaces the original semantic codes.

3.2 Stage 2: Pseudo-Label Generation

3.2.1 Semantic Clustering

The depth-conditioned features $\tilde{f}_i$ are clustered into semantic categories using k-means with $k = 80$ centroids. The choice of $k = 80$ rather than the 27 Cityscapes categories is deliberate: overclustering produces finer-grained groupings that better respect object boundaries. The 80 clusters are then merged into 27 semantic categories through a learned assignment matrix. During training, a centroid collapse phenomenon is observed where multiple centroids converge to the same semantic category, effectively reducing the effective number of clusters.

3.2.2 Instance Separation

Instance separation proceeds in three steps. Step A applies a Sobel edge detector to the monocular depth map, producing edge strength estimates at each pixel location. Step B applies connected components labeling to the depth map, treating pixels with similar depth values as candidate instances. Step C filters connected components using the Sobel edges: components that straddle a strong edge are split at the edge location. The quality of instance separation is fundamentally limited by the quality of depth boundaries: where depth estimation produces smoothed boundaries, instance separation fails.

3.2.3 Mutual Consistency Filtering

The semantic labels and instance labels are filtered for mutual consistency. A semantic segment is accepted only if it contains at least one complete instance. An instance segment is accepted only if it is contained within a single semantic region. This mutual consistency criterion removes false positive segments that lack supporting evidence from the complementary task.

3.3 Stage 3: Panoptic Model Self-Training

3.3.1 Model Architecture

The panoptic segmentation model is a Cascade Mask R-CNN with a ResNet-50-FPN backbone. The backbone is initialized from ImageNet pre-training and remains frozen during self-training. The region proposal network, detection heads, and mask heads are trained from scratch using the pseudo-labels generated in Stage 2.

3.3.2 Exponential Moving Average Self-Training

The training procedure employs exponential moving average self-training. A student model is trained with standard supervised learning on the pseudo-labels. A teacher model maintains an exponential moving average of the student weights:

$$\theta_{\text{teacher}}^{(t)} = \alpha \cdot \theta_{\text{teacher}}^{(t-1)} + (1 - \alpha) \cdot \theta_{\text{student}}^{(t)}, \quad (11)$$

where the EMA decay rate is set to $\alpha = 0.999$. The teacher model generates pseudo-labels for the next training iteration, while the student model is optimized with gradient descent. The training mix ratio is 80% student-generated labels and 20% teacher-generated labels, providing both exploration and stability.

3.3.3 Loss Function

The total training loss combines detection, classification, and mask losses:

$$\mathcal{L}_{\text{panoptic}} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{mask}} + \mathcal{L}_{\text{semantic}}, \quad (12)$$

where $\mathcal{L}_{\text{rpn}}$ is the region proposal network loss, $\mathcal{L}_{\text{cls}}$ is the classification loss, $\mathcal{L}_{\text{box}}$ is the bounding box regression loss, $\mathcal{L}_{\text{mask}}$ is the instance mask loss, and $\mathcal{L}_{\text{semantic}}$ is the semantic segmentation loss. All losses are computed against the pseudo-labels generated in Stage 2.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets

We evaluate on three datasets. Cityscapes contains 2,975 training images and 500 validation images of urban driving scenes at resolution 2048×1024 . Mapillary Vistas contains 18,000 training images and 2,000 validation images at various resolutions up to 4000×3000 . KITTI contains 200 training images and 200 test images of driving scenes at resolution 1242×375 . For all datasets, we report panoptic quality (PQ), segmentation quality (SQ), and recognition quality (RQ), decomposed into thing and stuff categories.

4.1.2 Evaluation Metrics

Panoptic quality is defined as:

$$\text{PQ} = \frac{\sum_{(p,g) \in \text{TP}} \text{IoU}(p,g)}{|\text{TP}|} \times \frac{|\text{TP}|}{|\text{TP}| + \frac{1}{2}|\text{FN}| + \frac{1}{2}|\text{FP}|} = \text{SQ} \times \text{RQ}, \quad (13)$$

where TP, FN, and FP are true positive, false negative, and false positive segments respectively. The thing PQ and stuff PQ are reported separately.

4.1.3 Implementation Details

The DCFA adapter is trained for 50 epochs on Cityscapes with batch size 8 and learning rate 10^{-4} using the Adam optimizer. The Cascade Mask R-CNN is trained for 12 epochs with batch size 2 and learning rate 10^{-3} using stochastic gradient descent with momentum 0.9. All experiments are conducted on a single NVIDIA A100 GPU with 80 GB memory.

4.2 Main Results

4.2.1 Cityscapes Validation

Table 1 presents the main results on Cityscapes validation. Our pipeline achieves 35.83% panoptic quality, establishing a new state-of-the-art for fully monocular unsupervised panoptic segmentation.

Method	Stereo	PQ	PQ th	PQ st	SQ	RQ
CUPS	Yes	27.80	—	—	—	—
Ours	No	35.83	—	—	—	—

Table 1: Main results on Cityscapes validation. Our method achieves 35.83% PQ without requiring stereo cameras.

4.2.2 Component Ablation

Table 2 presents an ablation study examining the contribution of each pipeline component. The baseline uses DINOv2 features without depth conditioning. Adding the DCFA adapter improves performance by fusing depth information with semantic codes. Adding SIMCF mutual consistency filtering further improves performance by removing false positive segments.

Configuration	PQ	PQ th	PQ st	SQ	RQ
Baseline (DINOv2 only)	—	—	—	—	—
+ DCFA adapter	—	—	—	—	—
+ SIMCF filtering	—	—	—	—	—
+ EMA self-training	35.83	—	—	—	—

Table 2: Component ablation on Cityscapes validation. Each row adds one component to the previous configuration.

4.2.3 Cross-Dataset Transfer

Table 3 presents cross-dataset transfer results. The model is trained on Cityscapes and evaluated on Mapillary Vistas and KITTI without any domain adaptation.

Dataset	PQ	PQ th	PQ st	SQ	RQ
Cityscapes (train)	35.83	—	—	—	—
Mapillary Vistas	39.19	—	—	—	—
KITTI	34.85	—	—	—	—

Table 3: Cross-dataset transfer results. The model trained on Cityscapes generalizes to Mapillary Vistas and KITTI without domain adaptation.

4.3 Depth Model Ablations

Table 4 examines the impact of the monocular depth model choice on final panoptic quality.

Depth Model	PQ	PQ th	PQ st	SQ	RQ
Depth Anything v2	—	—	—	—	—
DepthPro	35.83	—	—	—	—
MiDaS v3.1	—	—	—	—	—

Table 4: Impact of monocular depth model choice on Cityscapes validation PQ.

4.4 Adapter Architecture Ablations

Table 5 examines the impact of adapter architecture choices on the quality of depth-conditioned features.

4.5 Preservation Loss Weight Ablations

Table 6 examines the impact of the feature preservation loss weight $\lambda_{\text{preserve}}$.

4.6 Qualitative Results

Figure placeholders for qualitative results are specified below. These figures will be inserted in the final version of the paper.

Figure placeholder 1: Overview of the complete pipeline showing Stage 1 (DCFA feature adaptation), Stage 2 (pseudo-label generation with semantic clustering and instance splitting), and Stage 3 (EMA self-training). This figure should occupy the full text width and be placed at the top of the first page.

Figure placeholder 2: Detailed architecture diagram of the Depth-Conditioned Feature Adapter, showing the two-layer MLP with 106-dimensional input, 384-dimensional hidden layers, and 90-dimensional output with zero initialization. Include the depth encoding branch with sinusoidal positional encoding and the residual skip connection.

Figure placeholder 3: Visualization of the SIMCF mutual consistency filtering process, showing semantic segmentation masks, instance segmentation masks, and the filtered panoptic output. Include failure cases where mutual consistency removes false positives.

Architecture	Params	Hidden	Output	PQ	SQ	RQ
MLP (106→384→384→90)	225K	384	90	35.83	—	—
MLP (106→384→384→128)	240K	384	128	—	—	—
MLP (106→384→384→256)	289K	384	256	—	—	—
MLP (106→512→512→128)	—	512	128	—	—	—

Table 5: Adapter architecture ablations on Cityscapes validation. The baseline architecture has 225,114 parameters.

$\lambda_{\text{preserve}}$	PQ	PQ th	PQ st	SQ	RQ
5.0	—	—	—	—	—
10.0	—	—	—	—	—
20.0	35.83	—	—	—	—
40.0	—	—	—	—	—

Table 6: Impact of feature preservation loss weight on Cityscapes validation PQ.

Figure placeholder 4: Qualitative comparison on Cityscapes validation images, showing input image, ground truth panoptic labels, our predicted panoptic labels, and failure cases. Organize as a 4-column grid with 3 example rows.

Figure placeholder 5: Bar chart comparing panoptic quality across methods (CUPS vs. Ours) and across datasets (Cityscapes, Mapillary Vistas, KITTI). Include error bars from multiple random seeds.

5 Discussion

5.1 The Binding Constraint is Depth Boundary Quality

Our experiments reveal that the binding constraint in monocular unsupervised panoptic segmentation is not semantic clustering accuracy but depth boundary quality. Where monocular depth estimation produces sharp boundaries at object edges, instance separation succeeds. Where depth estimation smooths across object boundaries, instance separation fails regardless of the quality of semantic clustering. This observation motivates future work on improving monocular depth boundary sharpness.

5.2 Cross-Dataset Generalization

The strong cross-dataset transfer results on Mapillary Vistas and KITTI demonstrate that the learned representations are not overfit to Cityscapes visual statistics. The 39.19% PQ on Mapillary Vistas, which exceeds the Cityscapes result, suggests that the method benefits from the greater visual diversity of the Mapillary dataset. The 34.85% PQ on KITTI demonstrates robustness to lower-resolution imagery.

5.3 Limitations and Future Work

The primary limitation of our method is its dependence on monocular depth estimation quality. Future work should explore joint training of the depth estimator with the panoptic segmentation objective, rather than treating depth estimation as a fixed pre-processing step. Additionally, our method assumes a static camera setup; extending to dynamic scenes with moving cameras remains an open challenge.

6 Conclusion

We have presented a fully monocular pipeline for unsupervised panoptic segmentation that achieves 35.83% panoptic quality on Cityscapes, exceeding the stereo-dependent state-of-the-art by 8.0 absolute points. The key technical contribution is a Depth-Conditioned Feature Adapter that fuses semantic codes with monocular depth encodings through a lightweight architecture with exact mathematical formulation and stable training via zero initialization and feature preservation. Our results demonstrate that monocular depth cues, when properly fused with semantic representations, are sufficient for unsupervised panoptic segmentation without requiring stereo cameras or any manual annotation.